Ticket 07 Study Report: Dividend-Paying American Call Validation

SURF2026 American Option Risk Surfaces

Codex-assisted implementation report

June 16, 2026

Abstract

Ticket 07 validates the positive-dividend American call regime using the existing Ticket 04 American CN/PSOR solver without changing solver code. Unlike Ticket 06, where $q = 0$ and a standard American call should match the European call, this ticket sets $q = 0.08$ and checks whether the American solver can represent a regime where early exercise value may become meaningful. The validation compares American call values against Ticket 01 European Black-Scholes call values, checks the payoff obstacle $U \geq \Phi$, records PSOR convergence metadata, writes reproducible CSV tables, and produces three validation figures. This report is written for beginner researchers with accounting, data, machine-learning, and Python experience who are still learning American option pricing, dividends, obstacle problems, and CN/PSOR validation.

Contents

1 Role of Ticket 07 in the Whole Solver Project

The project is building a classical, validated American option benchmark before moving to boundary diagnostics, Greeks, stress regimes, datasets, or neural surrogate models. Ticket 07 is part of the validation ladder:

- Ticket 01 created payoff functions and European Black-Scholes closed-form utilities.
- Ticket 02 created uniform finite-difference grids, spatial operator coefficients, and simple boundary value helpers.
- Ticket 03 validated European Crank-Nicolson time stepping against closed-form prices.
- Ticket 04 implemented the American CN/PSOR obstacle core.
- Ticket 05 validated representative American put behavior.
- Ticket 06 validated the no-dividend American call control theorem, $q = 0$.

Ticket 07 changes the financial regime from $q = 0$ to $q > 0$. It asks whether the American exercise machinery can show meaningful American-minus-European value when dividends make early exercise more plausible. It does not modify the solver. It uses the existing solver as a black-box validation target and records whether the output is financially and numerically credible.

2 Theory and Methodology/Literature Connection

The documented source basis for this report is internal:

- `reports/01_literature/literature_map.md`;
- `reports/02_math/formulation_note.md`;
- `reports/03_solver/solver_validation_plan.md`;
- `reports/03_solver/solver_implementation_tickets.md`;
- the Ticket 01 through Ticket 06 reports and code.

The literature map and formulation note frame American options as obstacle problems or linear complementarity problems when early exercise is allowed. Ticket 04 implemented that LCP logic with PSOR. The solver validation plan then asks for staged validation: first European prices, then American puts, then no-dividend American calls, then dividend-paying American calls. Ticket 07 follows that sequence.

This reference integration pass adds verified citations for the external theory behind the project documents. Dividend-adjusted option pricing follows the Black-Scholes/Merton framework (Black and Scholes, 1973; Merton, 1973), and the American obstacle/LCP interpretation is connected to standard complementarity methodology (Cottle et al., 2009).

3 Why Dividends Can Make American Call Early Exercise Meaningful

A call option gives the holder the right to buy the underlying stock for strike K. The call payoff is

$$\Phi_{\mathrm{call}}(S) = \max(S - K, 0). \tag{1}$$

For a no-dividend stock, exercising a call early is usually unattractive. The holder gives up remaining optionality and pays the strike early without receiving any dividend benefit. That is why Ticket 06 checked the theorem

$$C_{\text{American}}(S,T;q=0) \approx C_{\text{European}}(S,T;q=0). \tag{2}$$

With a continuous dividend yield $q > 0$, the trade-off changes. Holding the option does not receive the underlying dividend stream. Exercising the call gives ownership of the stock, and stock ownership benefits from dividends. When the dividend benefit is large enough, especially for sufficiently in-the-money calls, early exercise can become economically meaningful.

This does not mean every positive-dividend American call must exercise early under every parameter set. It means the solver should be able to represent the possibility. Ticket 07 therefore checks for a positive American-minus-European value in a representative $q = 0.08$ case, while still avoiding any free-boundary claim.

4 Contrast with Ticket 06 No-Dividend Call Validation

Ticket 06 was a control test: if $q = 0$, the American call should not be meaningfully above the European call. Ticket 07 is the positive-dividend test: if $q > 0$, the American call should still dominate payoff and the European call, and it may be meaningfully above the European call.

Ticket	Financial setting	Main validation idea
Ticket 06	No-dividend American call, $q = 0$	American value should approximately equal European value; no false early-exercise premium.
Ticket 07	Dividend-paying American call, $q = 0.08$	American value should dominate payoff and European value; positive American-minus-European value can appear.

This contrast is important for beginner researchers. The same American solver can be tested in a case where early exercise should not matter and then in a case where it can matter. Passing both checks gives stronger evidence than looking at only one regime.

5 Dividend-Paying American Call Financial Intuition

For a European call with continuous dividends, the closed-form benchmark includes the dividend discount factor e^{-qT}. Intuitively, future stock value is reduced by the dividend yield in the risk-neutral pricing formula:

$$C_{\text{European}} = Se^{-qT}N(d_1) - Ke^{-rT}N(d_2). \tag{3}$$

For an American call, the holder can also compare:

- the continuation value: keep the option alive; and
- the immediate exercise value: take $\max(S-K,0)$ now.

The American obstacle condition is

$$U(S,\tau) \geq \Phi_{\text{call}}(S). \tag{4}$$

When $U > \Phi$, continuation is better at that grid node. When $U = \Phi$, immediate exercise is competitive with continuation. Ticket 07 reports the obstacle gap $U - \Phi$, but only as a validation diagnostic. It does not infer an exercise boundary from that gap.

6 Validation Criteria

6.1 Positive-Dividend Setup

Every Ticket 07 case must use a call option and $q > 0$. The default cases use:

$$K = 1, \quad T = 1, \quad r = 0.05, \quad q = 0.08, \quad \sigma = 0.20, \quad S_{\max} = 4.$$

The tests also verify that every validation and selected-spot row reports a positive q.

6.2 Obstacle Condition

The American call value should never fall below payoff:

$$U(S_i,\tau_n) - \Phi(S_i) \geq 0. \tag{5}$$

The experiment reports the minimum obstacle gap and the maximum obstacle violation over the full stored value grid. A zero maximum obstacle violation means the PSOR projection respected the obstacle on the tested grids.

6.3 American Versus European Call

An American option gives at least as much exercise flexibility as the corresponding European option. In the positive-dividend call case, the experiment checks:

$$U_{\text{American call}}(S,T;q>0) \geq C_{\text{European call}}(S,T;q>0), \tag{6}$$

up to numerical tolerance.

6.4 Dividend-Call Early-Exercise Value Diagnostic

The diagnostic difference is

$$\Delta_{\text{Am}-\text{Eur}}(S) = U_{\text{American call}}(S,T) - C_{\text{European call}}(S,T). \tag{7}$$

Ticket 07 reports the minimum, maximum, maximum absolute difference, RMSE, and selected-spot values. Positive $\Delta_{\text{Am}-\text{Eur}}$ is evidence that the solver is representing the dividend-call early-exercise regime. It is not a boundary location.

6.5 Target Region

Summary error and difference metrics focus on:

$$0.4 \leq S/K \leq 1.8.$$

This follows the existing solver validation plan. The selected-spot table additionally includes $S/K = 2.0$ because the dividend-call effect is particularly visible for higher spot values.

7 What Ticket 07 Is Not

Ticket 07 does not redo the no-dividend call theorem validation from Ticket 06. It uses $q > 0$ only.

Ticket 07 is also not boundary extraction. It does not compute a continuation premium module, does not search for a free-boundary location, and does not label any figure as a boundary. The American-minus-European curve is a pricing comparison, not a free-boundary diagnostic.

Finally, Ticket 07 does not implement Greeks, stress maps, datasets, or neural-network files. Those require a more complete validated solver pipeline and belong to later work.

8 Implementation Design Decisions

Decision	Reason
Use the Ticket 04 solver unchanged	Ticket 07 is a validation ticket. Changing the solver would mix implementation and validation concerns.
Use $q = 0.08$	The dividend yield is positive and large enough to make American-minus-European value visible in the representative case.
Use medium and fine grids	The 120×120 and 180×180 cases give a light refinement audit without becoming a full grid/domain sensitivity study.
Use nearest-grid selected spots	Nearest-grid reporting is simple and reproducible for beginner readers. The report states that this is not interpolated convergence analysis.
Generate three figures	The value comparison, American-minus-European curve, and PSOR iteration curve each support a specific validation question.
Avoid obstacle-gap figure	Obstacle gaps are reported in tables only to reduce the chance of confusing this ticket with boundary extraction.

9 Code-Level Function Explanations

Function	Role
`run_validation_cases`	Runs the medium and fine positive-dividend American call cases, then builds validation rows, selected-spot rows, and figure metadata.
`summarize_dividend_call_case`	Computes American-versus-European metrics, obstacle diagnostics, positive-difference node count, and PSOR convergence summaries for one solver result.
`selected_spot_rows`	Reports payoff, European call, American call, American-minus-European, and American-minus-payoff at selected moneyness points.
`nearest_spot_index`	Chooses the nearest grid node to a requested spot so selected-spot reporting is reproducible.

Function	Role
`write_csv`	Writes stable CSV artifacts with fixed column ordering.
`create_value_comparison_figure`	Creates the American call, European call, and payoff comparison figure if matplotlib is available.
`create_american_minus_european_figure`	Creates the American-minus-European curve for the dividend-call value diagnostic.
`create_psor_iterations_figure`	Creates the PSOR iteration-count curve by time step to summarize convergence behavior.
`main`	Runs the full Ticket 07 experiment and writes CSV and figure artifacts.

10 Validation Result Tables and Interpretation

10.1 Validation Cases and Parameters

Case	K	T	r	q	σ	M	N	$S_{\max}$
Medium	1.0	1.0	0.05	0.08	0.20	120	120	4.0
Fine	1.0	1.0	0.05	0.08	0.20	180	180	4.0

Both validation cases use $q = 0.08 > 0$. This distinguishes Ticket 07 from the no-dividend Ticket 06 control case.

10.2 American-Versus-European Summary

Case	Max abs diff	RMSE	Min Am. - Eur.	Max Am. - Eur.	Max diff spot	Positive nodes
Medium	8.9422×10^{-2}	$4.2654{\times}10^{-2}$	4.1188×10^{-8}	8.9422×10^{-2}	1.8	43
Fine	8.9422×10^{-2}	$4.2454{\times}10^{-2}$	1.6879×10^{-8}	8.9422×10^{-2}	1.8	64

The American-minus-European difference is nonnegative in the target region within the reported tolerance and is materially positive in the representative dividend case. The maximum target region difference is about 0.0894, and the selected $S/K = 2.0$ row shows a difference of about 0.1050. This is the key financial evidence in Ticket 07: the solver does not collapse the positive-dividend call case into the no-dividend theorem behavior.

10.3 Obstacle Summary

Case	Min obstacle gap	Max obstacle violation
Medium	0	0
Fine	0	0

The obstacle diagnostics show no violation in the stored value grids. This confirms that the American call values were projected above payoff during PSOR time stepping. Obstacle validity is necessary, but it is not sufficient by itself. Ticket 07 also needs the European comparison and PSOR convergence evidence.

10.4 PSOR Convergence Metadata

Case	All steps converged	Step count	Max iterations	Mean iterations	Max final update
Medium	True	120	10	7.35	9.4309×10^{-9}
Fine	True	180	11	7.1111	9.9674×10^{-9}

Every PSOR step converged in both cases. The maximum final update remained below the intended 10^{-8} tolerance scale. This makes the American-versus-European comparison more credible: the reported dividend-call premium is not being produced by visible PSOR non-convergence.

10.5 Selected Spot Price Comparison

Case	S/K	Payoff	European call	American call	Am. - Eur.	Am. - payoff
Medium	0.5	0	0.00000484	0.00000633	0.00000149	0.00000633
Medium	0.8	0	0.00817709	0.00836716	0.00019007	0.00836716
Medium	1.0	0	0.06142998	0.06513118	0.00370119	0.06513118
Medium	1.2	0.20000000	0.18282650	0.20369502	0.02086852	0.00369502
Medium	1.5	0.50000000	0.43612545	0.50000000	0.06387455	0
Medium	2.0	1.00000000	0.89503490	1.00000000	0.10496510	0
Fine	0.488889	0	0.00000301	0.00000352	0.00000051	0.00000352
Fine	0.8	0	0.00817709	0.00842129	0.00024420	0.00842129
Fine	1.0	0	0.06142998	0.06529167	0.00386168	0.06529167
Fine	1.2	0.20000000	0.18282650	0.20376496	0.02093846	0.00376496
Fine	1.488889	0.48888889	0.42612581	0.48888889	0.06276307	0
Fine	2.0	1.00000000	0.89503490	1.00000000	0.10496510	0

The selected-spot rows make the dividend-call effect readable. Near $S/K = 1.0$, the American call is modestly above the European call. At $S/K = 1.5$ and $S/K = 2.0$, the American value is at payoff in the final-time slice and is meaningfully above the European value. The report does not use these rows to infer an exercise boundary; they are selected price checks only.

10.6 Medium-to-Fine Comparison

The largest selected-spot medium-to-fine value difference reported by the experiment is 0.0111111. This is a nearest-grid audit, not an interpolated convergence proof. It is driven mainly by selected spots where one grid lands exactly on a payoff node and the other grid lands on the nearest neighboring node. A full grid and domain sensitivity study belongs to a later ticket.

11 Figure Interpretation

Figure 1 plots the fine-grid American call value, European call value, and payoff over moneyness. The American curve is above the European curve in the dividend-paying case, especially for higher spot values. The payoff curve provides the lower obstacle that the American value must respect.

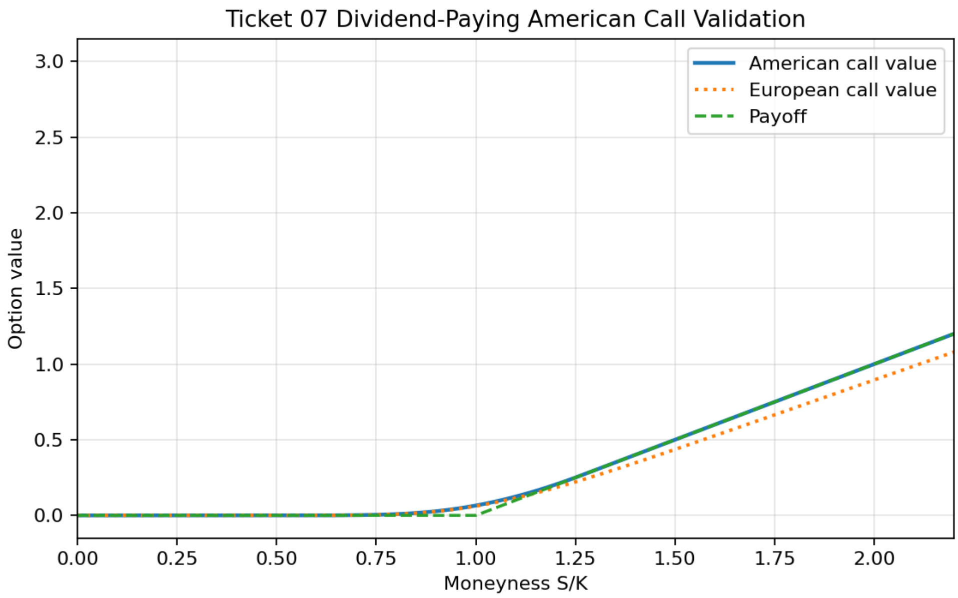

Figure 1: Ticket 07 dividend-paying American call value, European call value, and payoff comparison.

Figure 2 plots American call value minus European call value. This directly shows the positive-dividend value diagnostic. The zero line is included for reference. No boundary is extracted or labeled.

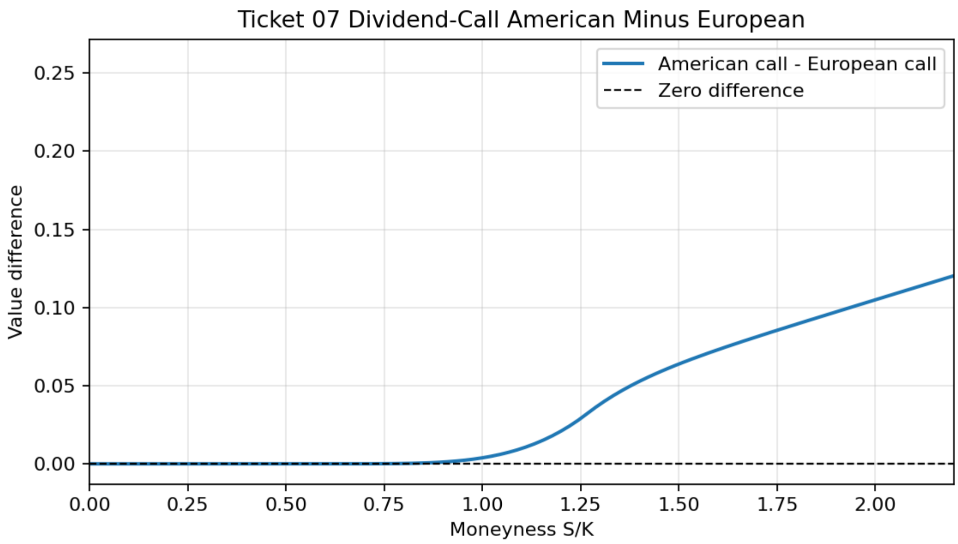

Figure 2: Ticket 07 American-minus-European value diagnostic for the dividend-paying call.

Figure 3 shows the number of PSOR iterations used at each time step in the fine case. The curve is a convergence-behavior diagnostic: it helps show that the validation prices come from converged LCP solves, not hidden failed iterations.

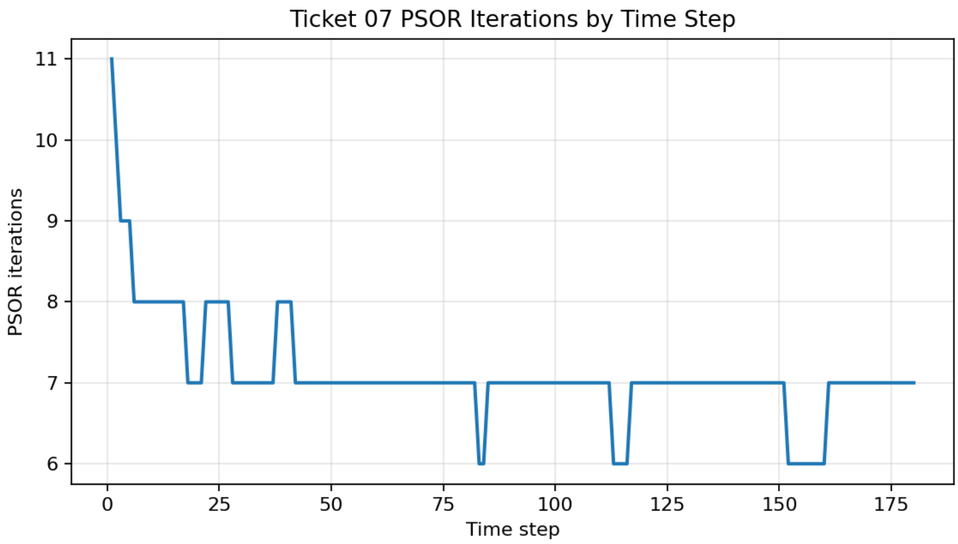

Figure 3: Ticket 07 PSOR iteration counts by time step for the fine positive-dividend call case.

12 Testing Strategy

Test category	What it proves	What it does not prove
Script/artifact tests	The experiment runs and writes the expected Ticket 07 CSV and PNG artifacts.	Full validation across all dividend regimes.
CSV schema tests	The validation and selected-spot tables have stable columns.	That every later diagnostic has already been designed.
$q > 0$ tests	Every validation row is genuinely positive-dividend.	Sensitivity to different dividend yields.
Obstacle tests	American call values dominate payoff within tolerance.	That a free boundary has been found.
European comparison tests	American values dominate European values within tolerance.	A formal convergence proof.
Positive-difference tests	The representative dividend case shows visible American-minus-European value.	That all positive-dividend calls must exercise early.
PSOR metadata tests	Default time steps converge and expose iteration/update summaries.	Convergence under every possible grid or relaxation parameter.
Nearest-grid test	Selected-spot reporting is predictable.	Interpolation accuracy.
Figure tests	Meaningful PNGs are created when matplotlib is available.	That plots alone validate the method.

Test category	What it proves	What it does not prove
Scope and docstring tests	Ticket 07 files identify their ticket and do not introduce forbidden APIs/files.	Future tickets that intentionally add diagnostics.

13 Test Results and Interpretation

The final test command was:

```
PYTHONPYCACHEPREFIX=/private/tmp/codex_pycache PYTHONPATH=src MPLCONFIGDIR=/private/
    tmp/matplotlib-cache FC_CACHEDIR=/private/tmp/fontconfig-cache python3 -m
    unittest discover -s tests
```

Observed result during Ticket 07 implementation:

```
Ran 76 tests

OK
```

The experiment command was:

```
PYTHONPYCACHEPREFIX=/private/tmp/codex_pycache PYTHONPATH=src MPLCONFIGDIR=/private/
    tmp/matplotlib-cache FC_CACHEDIR=/private/tmp/fontconfig-cache python3
    experiments/04_dividend_american_call_validation.py
```

The experiment wrote two validation rows, twelve selected-spot rows, and three PNG figures.

14 Implementation Pitfalls and Debugging Notes

Recorded item or pitfall	Symptom	Fix or guard	Beginner lesson
Expected TDD red run	The new tests failed with `FileNotFoundError` because the Ticket 07 experiment script did not exist yet.	Create `experiments/04_dividend_american_call_validation.py`.	A red test confirms the test is aimed at the new artifact.
Accidentally using $q = 0$	The experiment would repeat Ticket 06 rather than testing the dividend regime.	Validate every case and output row has $q > 0$.	The theorem regime is controlled by assumptions, not by filenames.
Overclaiming dividend behavior	A positive dividend yield does not guarantee early exercise under every parameter set.	Phrase results as evidence for this representative $q = 0.08$ case.	Validation evidence is parameter-specific unless a broader study is run.

Recorded item or pitfall	Symptom	Fix or guard	Beginner lesson
Confusing premium diagnostics with boundary extraction	American-minus-European and obstacle gap curves can tempt readers to infer an exercise boundary.	Report pricing differences and state that no boundary is extracted or labeled.	Diagnostics and boundary algorithms are different tasks.
Checking only $U \geq \Phi$	The solver could respect payoff but still fail the European comparison.	Include American-versus-European rows, metrics, and figures.	American validation needs multiple consistency checks.
Ignoring PSOR metadata	Positive premiums could be suspect if time steps did not converge.	Report convergence flags, iteration counts, and final update values.	Numerical evidence should include solver-status evidence.
Creating too many figures	Extra figures can distract from the validation question.	Use three figures, each with a distinct validation purpose.	Every plot should answer a specific question.

No unexpected solver bugs were recorded during this ticket. The only failed test iteration was the planned red TDD run before the Ticket 07 experiment script existed.

15 Limitations

Not included in Ticket 07	Reason
No-dividend call validation	Already covered in Ticket 06.
American put validation	Already covered in Ticket 05.
Free-boundary extraction	Ticket 07 does not infer or label exercise boundaries.
Continuation-premium module	Boundary-oriented diagnostics are later work.
Greeks	Delta and Gamma diagnostics are later work.
Stress maps, datasets, neural networks	These require a more complete validated solver pipeline.
Full grid/domain sensitivity	The current medium-to-fine comparison is preliminary only.
Broad dividend sensitivity	The ticket uses one representative positive-dividend setup, $q = 0.08$.

16 Lessons Learned / Study Notes

The first lesson is that dividend yield changes American call economics. With $q = 0$, early exercise should not be valuable for a standard call. With $q > 0$, early exercise can become meaningful because stock ownership receives dividend benefits that option ownership does not.

The second lesson is that American-minus-European value is a pricing diagnostic, not automatically a boundary. It can show that the American exercise right has value, but boundary extraction requires separate logic and careful definitions.

The third lesson is that $U \geq \Phi$, American ≥ European, and PSOR convergence answer different questions. A credible validation ticket should check all three.

The fourth lesson is that selected-spot tables are for human inspection. They help students see what is happening at familiar moneyness levels, but they do not replace full convergence or sensitivity analysis.

17 Link to the Next Ticket

Ticket 07 completes the basic American call validation pair:

- Ticket 06 showed the no-dividend control behavior: American call approximately equals European call when $q = 0$.
- Ticket 07 showed positive-dividend behavior: American call can be meaningfully above the European call while respecting payoff and PSOR convergence checks.

The next ticket can build on this validated solver ladder to add a new diagnostic layer. Later work still needs boundary extraction, complementarity diagnostics, Greeks, grid/domain sensitivity, stress maps, datasets, and neural surrogate work. Those are deliberately outside Ticket 07.

18 Reproducibility Record

Item	Record
Experiment script	`experiments/04_dividend_american_call_validation.py`.
Tests	`tests/test_dividend_american_call_validation.py`.
Validation CSV	`results/01_solver_validation/tables/ticket_07_dividend_american_call_validation.csv`.
Selected-spots CSV	`results/01_solver_validation/tables/ticket_07_dividend_american_call_selected_spots.csv`.
Figures	`results/01_solver_validation/figures/ticket_07_dividend_american_call_value_comparison.png`; `results/01_solver_validation/figures/ticket_07_dividend_american_call_american_minus_european.png`; `results/01_solver_validation/figures/ticket_07_dividend_american_call_psor_iterations.png`.
Report source	`reports/03_solver/tickets/ticket_07_dividend_american_call_validation.tex`.
Report PDF	`reports/03_solver/tickets/ticket_07_dividend_american_call_validation.pdf`.

Commands used:

```
PYTHONPYCACHEPREFIX=/private/tmp/codex_pycache PYTHONPATH=src MPLCONFIGDIR=/private/
    tmp/matplotlib-cache FC_CACHEDIR=/private/tmp/fontconfig-cache python3
    experiments/04_dividend_american_call_validation.py
```

```
PYTHONPYCACHEPREFIX=/private/tmp/codex_pycache PYTHONPATH=src MPLCONFIGDIR=/private/
    tmp/matplotlib-cache FC_CACHEDIR=/private/tmp/fontconfig-cache python3 -m
    unittest discover -s tests
PYTHONPYCACHEPREFIX=/private/tmp/codex_pycache PYTHONPATH=src python3 -m compileall
    -q src tests experiments
```

References

Fischer Black and Myron Scholes. The pricing of options and corporate liabilities. *Journal of Political Economy*, 81(3):637–654, 1973. doi: 10.1086/260062.

Richard W. Cottle, Jong-Shi Pang, and Richard E. Stone. *The Linear Complementarity Problem*. Society for Industrial and Applied Mathematics, 2009. ISBN 9780898716863. doi: 10.1137/1.9780898719000.

Robert C. Merton. Theory of rational option pricing. *The Bell Journal of Economics and Management Science*, 4(1):141, 1973. doi: 10.2307/3003143.